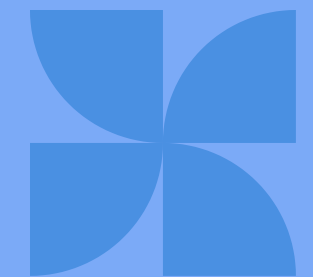




Wrocław University
of Science and Technology

Zero-Knowledge Proofs



Presented by

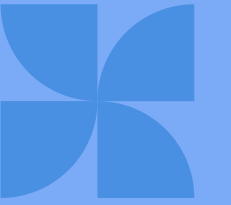
Belliot Thomas - 294623

Guedelha Nunes João - 294658

Outline of the presentation



- 1. Context Introduction**
- 2. Core idea & model**
- 3. Live Demo**
- 4. Discussion**



1985

→ Introduction to Zero-knowledge proofs in
The knowledge complexity of interactive proof-systems

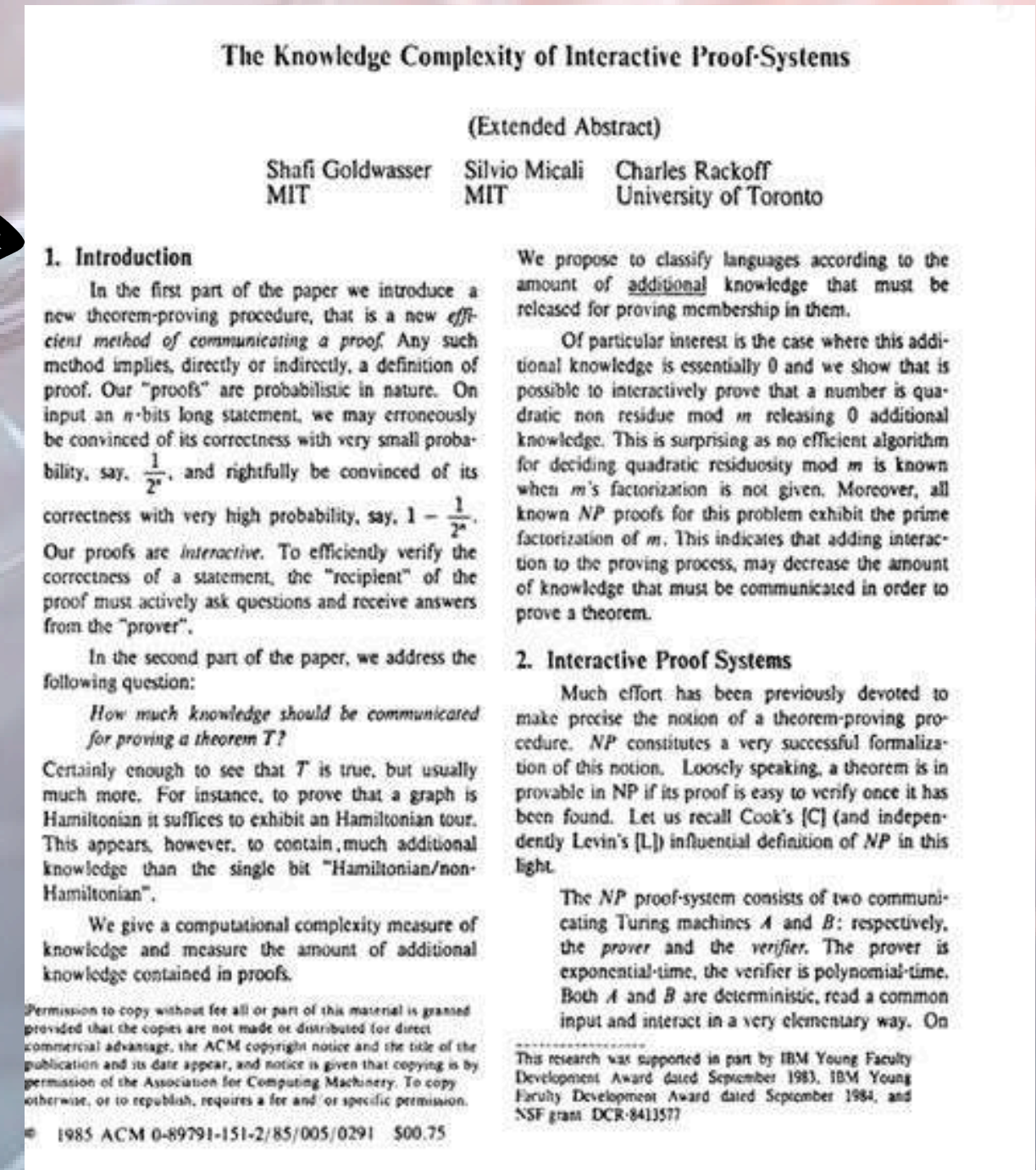
→ Concept of **knowledge complexity** : a measurement of
the amount of knowledge transferred

→ *Gödel Prize* (annual prize for outstanding papers in the
area of theoretical computer science) in 1993

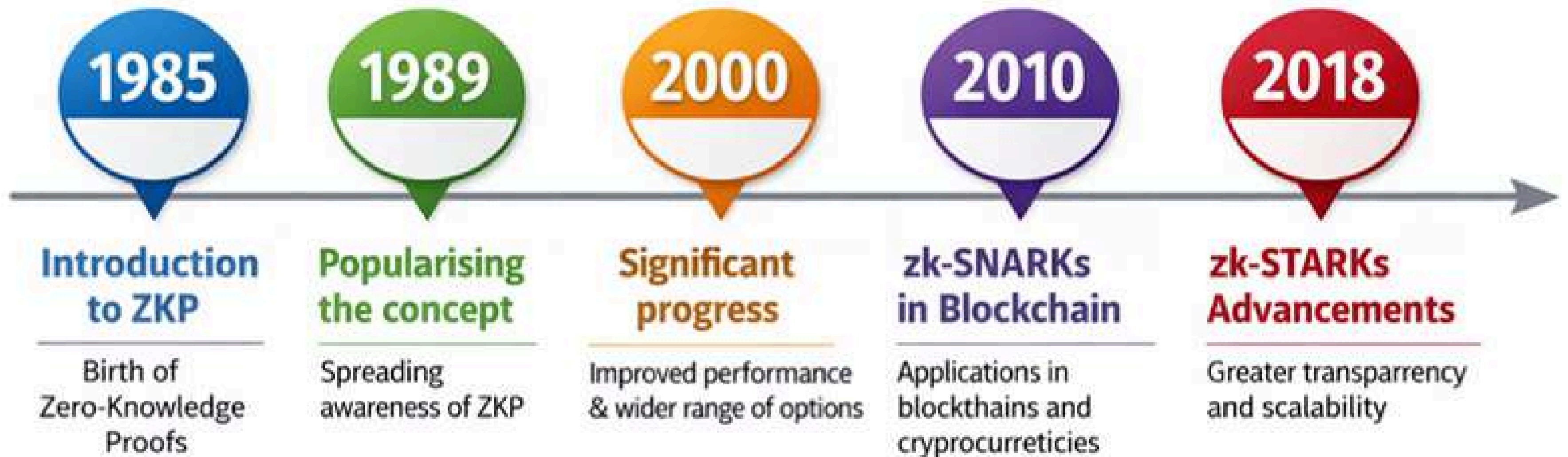
1989

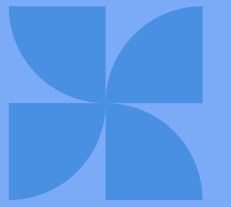
→ Popularising the concept among the general public :
How to Explain Zero-Knowledge Protocols to Your Children

(Wikipedia)



by Shafi Goldwasser, Silvio Micali
and Charles Rackoff

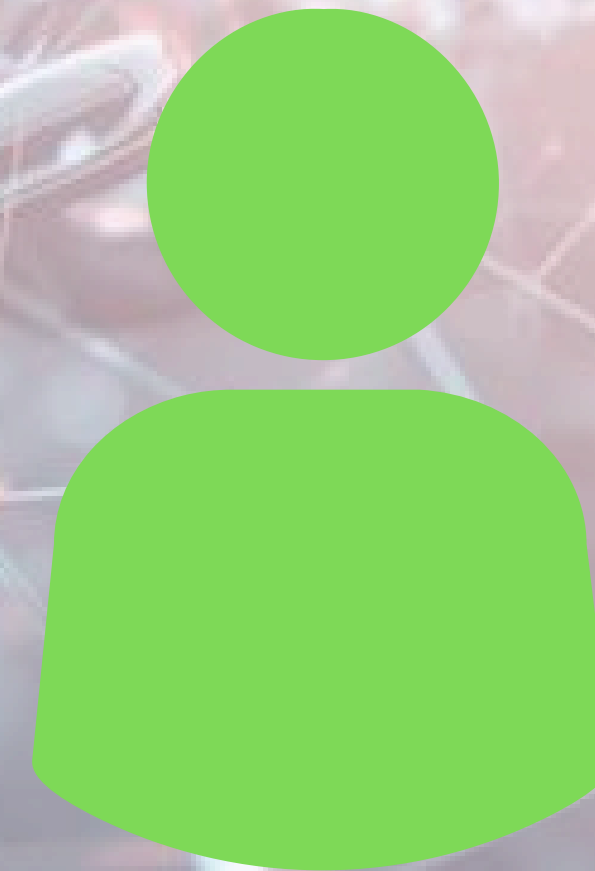




Prover's
secret



Prover

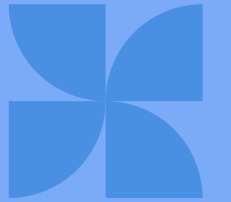


Verifier

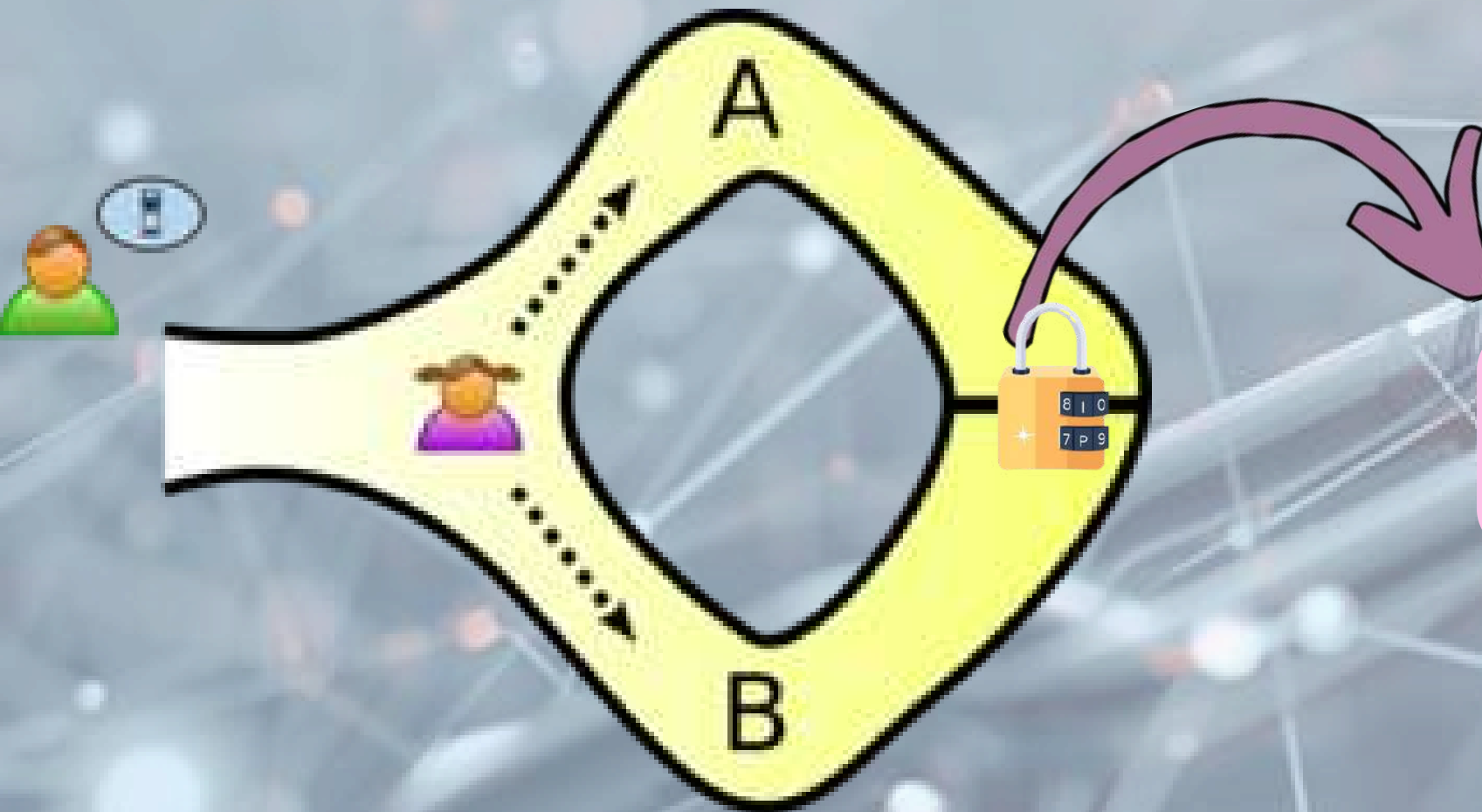
Objective: the **prover demonstrates that they know their secret without revealing it**

Context Introduction

Ex. 1 – The Alibaba cave



Source: Jean-Jacques Quisquater , *How to Explain Zero-Knowledge Protocols to Your Children*

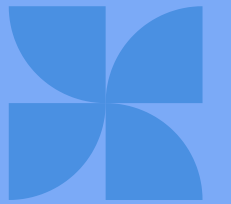


Door that opens with a password

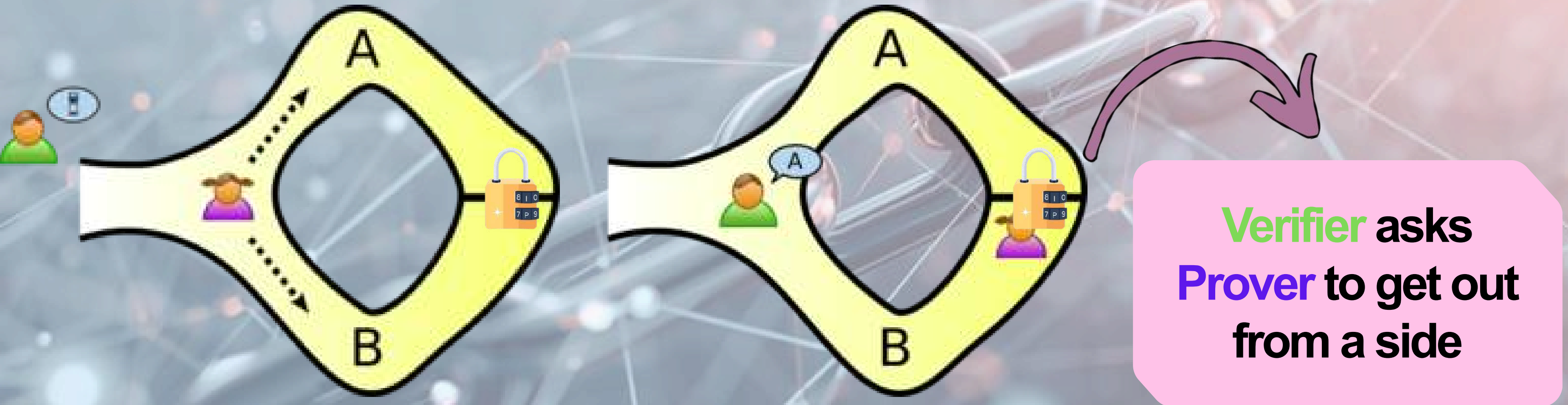
Objective: the **prover demonstrates that they know their secret without revealing it**

Context Introduction

Ex. 1 – The Alibaba cave



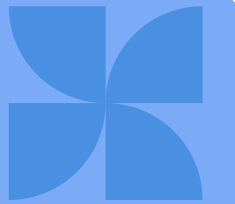
Source: Jean-Jacques Quisquater , *How to Explain Zero-Knowledge Protocols to Your Children*



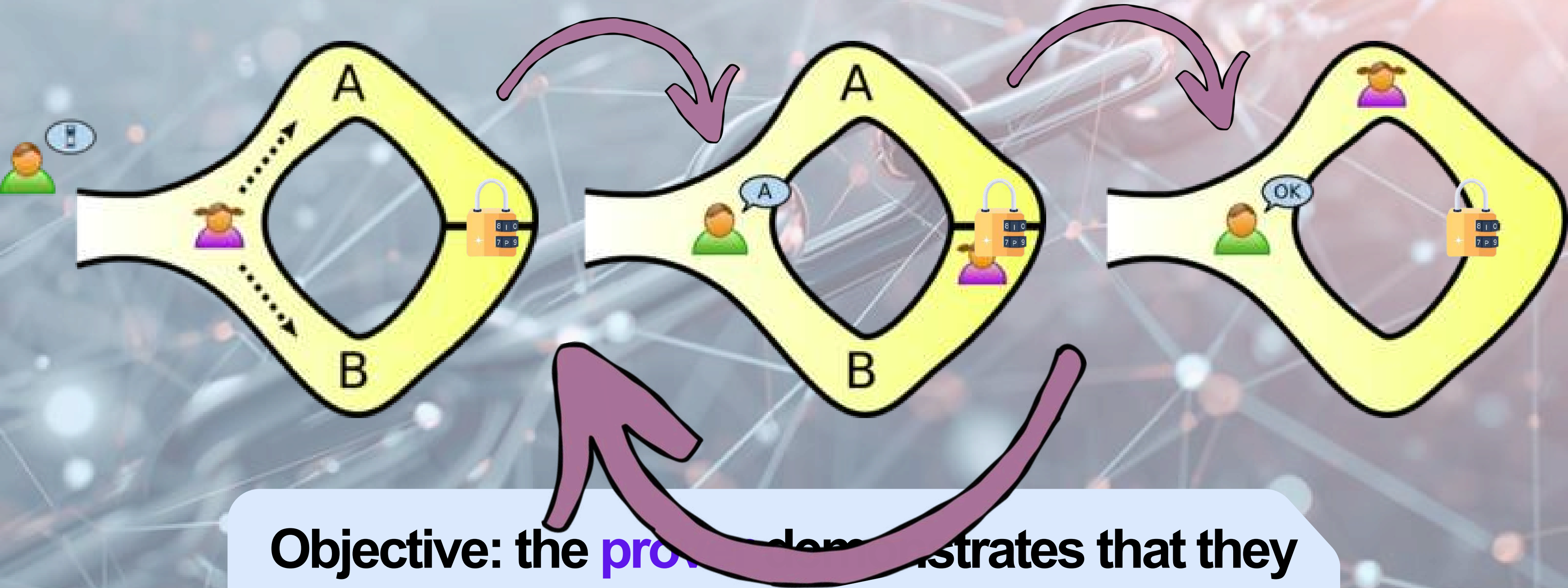
Objective: the **prover** demonstrates that they know their secret without revealing it

Context Introduction

Ex. 1 – The Alibaba cave



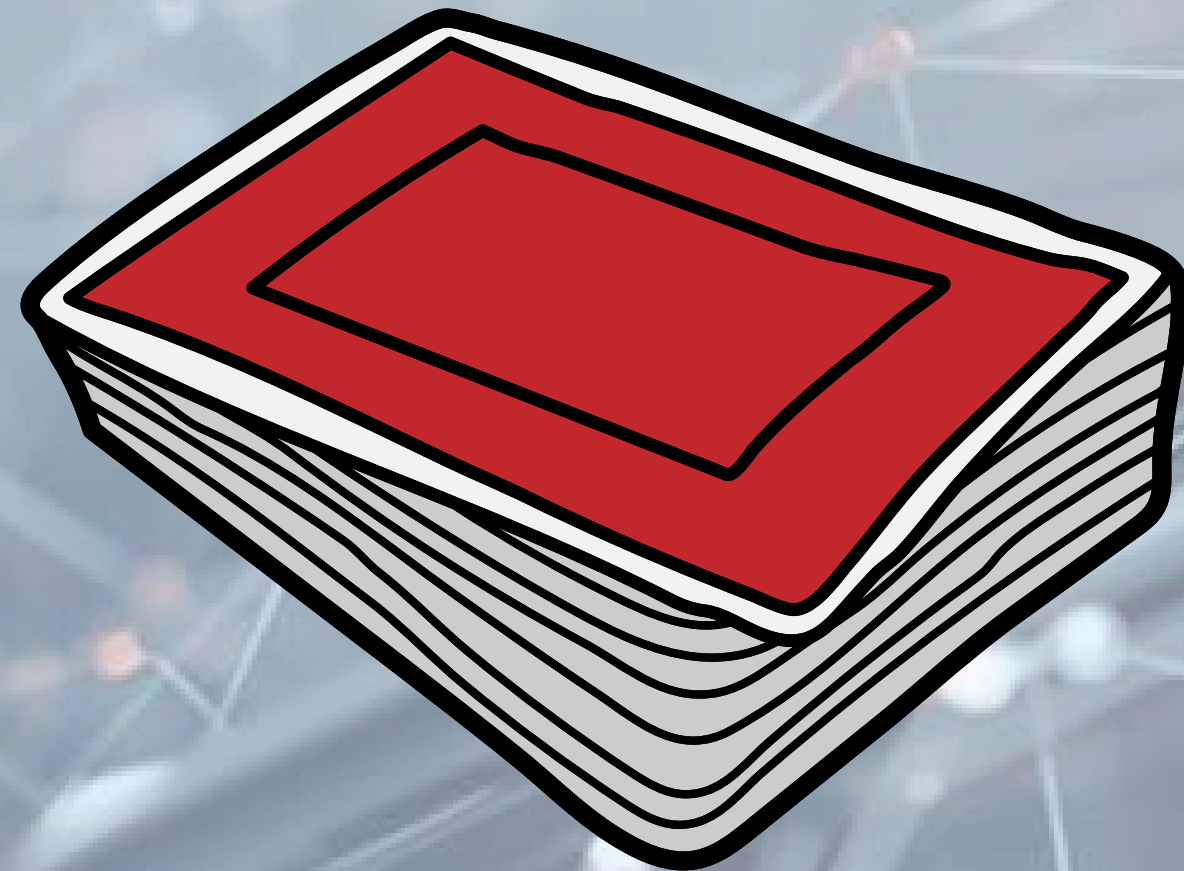
Source: Jean-Jacques Quisquater, *How to Explain Zero-Knowledge Protocols to Your Children (1990)*



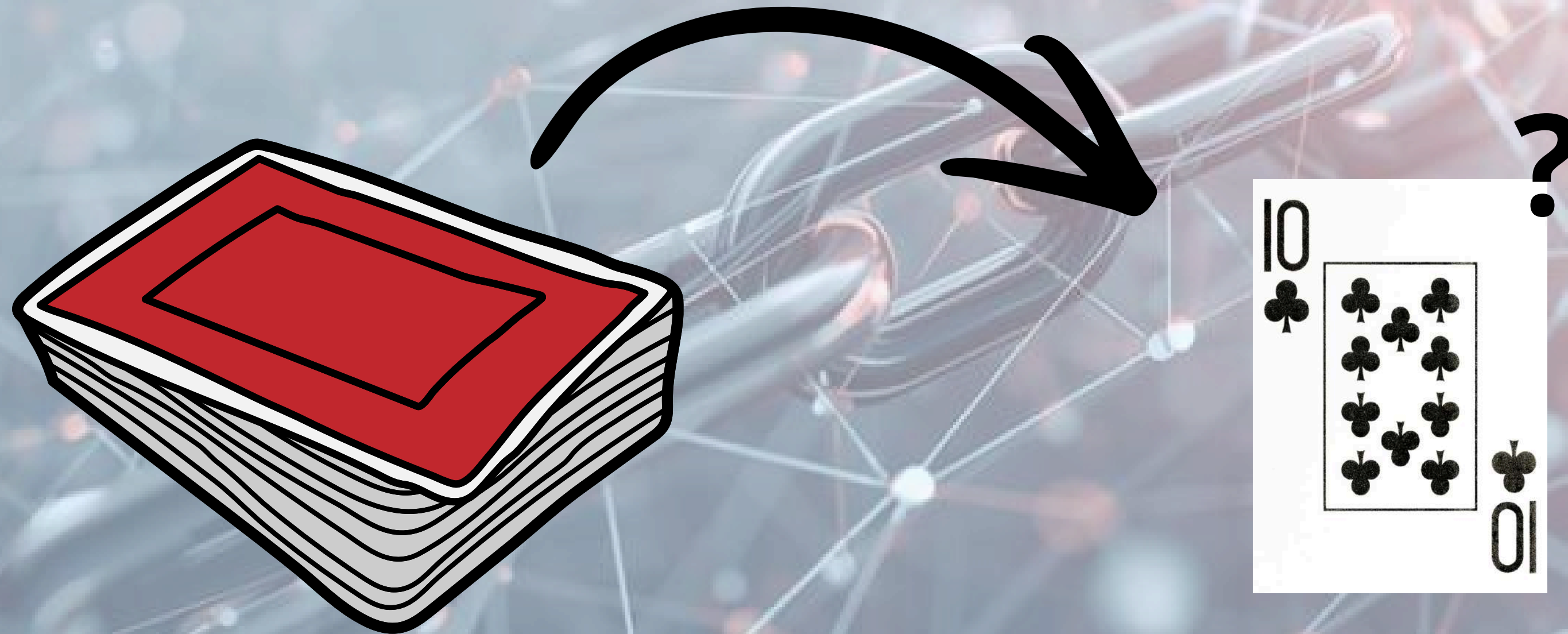
Objective: the prover demonstrates that they know their secret without revealing it



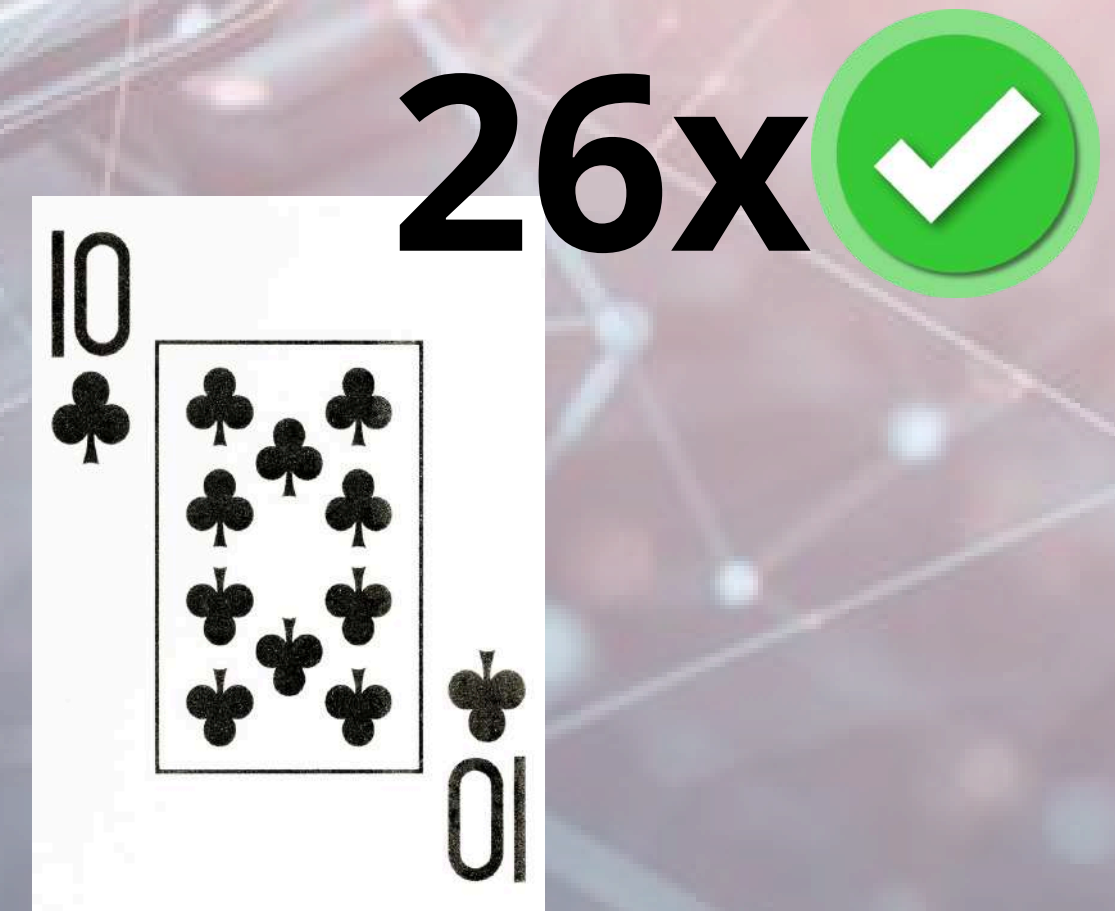
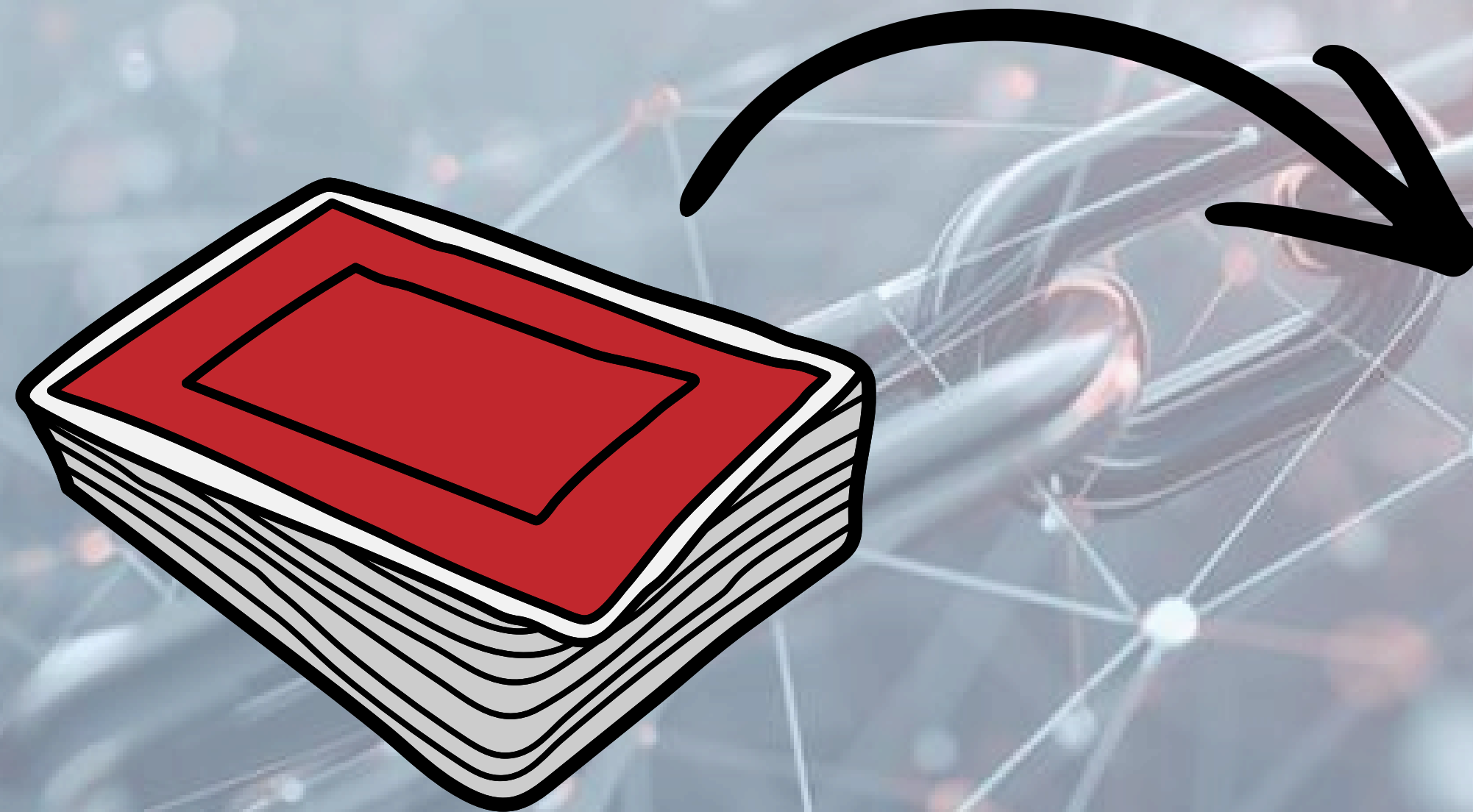
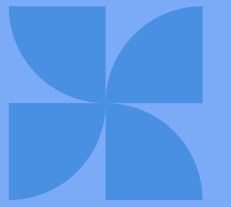
Secret: the prover knows his card color



Step 1: The **prover takes one card out of the deck without revealing it.**



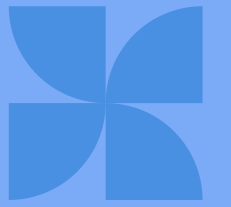
Step 2: The **prover shows all the black cards that are in the deck.**



Step 3: If the **prover** shows 26 black cards, he just convinced the **verifier**.

Context Introduction

Ex. 3 - The Sudoku grid

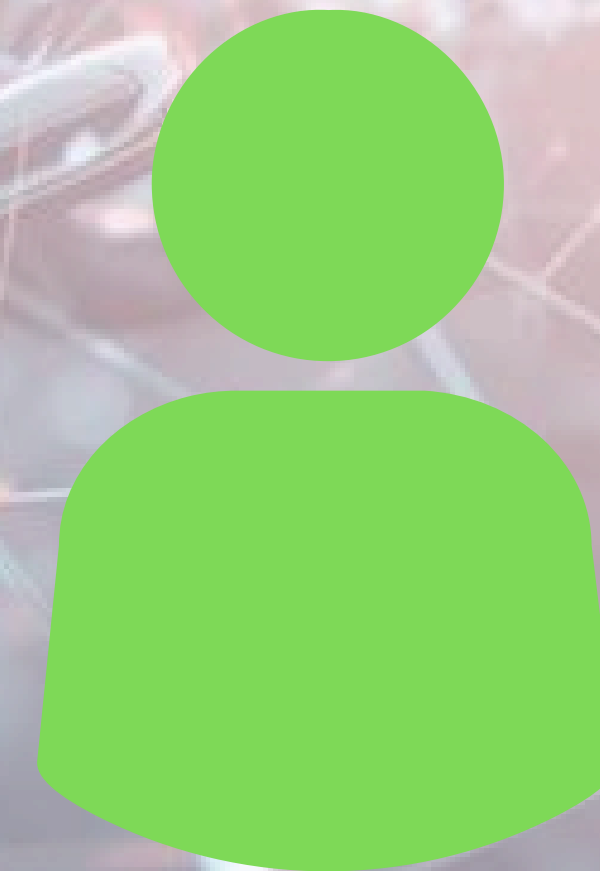


Prover's
secret

2	9	4	1	7	3	5	8	6
1	5	6	2	8	9	3	4	7
3	8	7	4	6	5	1	9	2
5	7	1	3	9	2	4	6	8
4	2	3	6	1	8	7	5	9
8	6	9	5	4	7	2	3	1
9	4	2	8	5	1	6	7	3
6	1	8	7	3	4	9	2	5
7	3	5	9	2	6	8	1	4



Prover

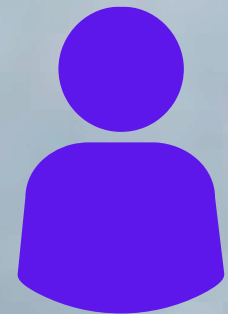
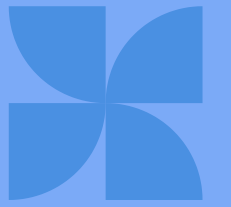


Verifier

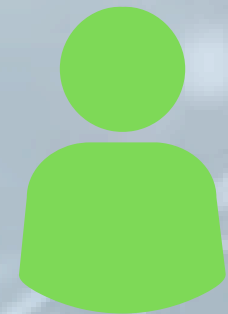
Objective: the **prover demonstrates that they know their secret without revealing it**

Context Introduction

Ex. 3 - The Sudoku grid



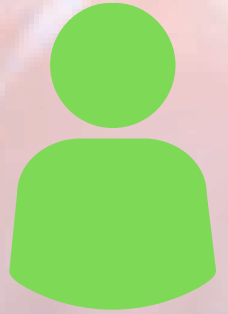
Cut out the grid,
turn all the pieces
of paper face
down, except for
the starting
numbers on the
grid



Select any row,
column or 3x3
square



Takes the 9
corresponding
squares of paper,
shuffles them, and
hands them to
Verifier



Checks that
the numbers
1 to 9 are
present



Objective: the **prover demonstrates that they
know their secret without revealing it**

Context Introduction

Why are ZKP important ?



Blockchain

Improve the confidentiality
and anonymity of
transactions



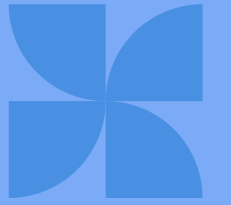
Cryptocurrencies

Improve the confidentiality
and anonymity of
transactions



Nuclear disarmament

Determine whether an object is a
nuclear weapon without revealing
sensitive information about its
internal structure



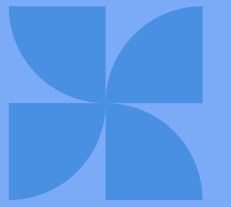
Zero-Knowledge Proofs rest on **3 foundational properties** :

- Completeness: Statement to be proven is true && both the prover and the verifier follow the protocol honestly → the verifier will be convinced with overwhelming probability.
- Soundness: statement false → cheating prover has a negligible probability of convincing the verifier.
- Zero-Knowledge: The verifier learns nothing new about the secret.

Source:



Mathematical Definition of Zero-Knowledge Proofs: Concepts and Examples
(by Daisuke Ishii ; 2025)



Let $L \subseteq \{0, 1\}^*$ be a language.

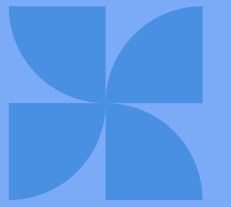
Zero-Knowledge Proof for L : interactive proof system (P, V) between **prover P** & **verifier V** such that:

- **Completeness**: For any $x \in L$, if both **P** and **V** follow the protocol correctly, $\Pr[\text{Verifier accepts}] \geq 1 - \epsilon(n)$, where $\epsilon(n) = o(1/n)$

Source:



Mathematical Definition of Zero-Knowledge Proofs: Concepts and Examples
(by Daisuke Ishii ; 2025)



Let $L \subseteq \{0, 1\}^*$ be a language.

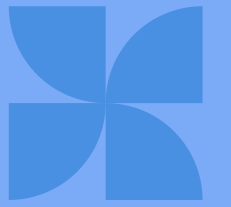
Zero-Knowledge Proof for L : interactive proof system (P, V) between **prover P** & **verifier V** such that:

- **Completeness**
- **Soundness:** For any $x \notin L$ and for any (possibly cheating) prover P ,
 $\Pr[\text{Verifier accepts}] \leq \varepsilon(n)$, where
 $\varepsilon(n) = o(1/n)$.

Source:



Mathematical Definition of Zero-Knowledge Proofs: Concepts and Examples
(by Daisuke Ishii ; 2025)



Let $L \subseteq \{0, 1\}^*$ be a language.

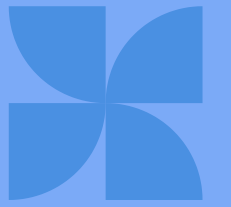
Zero-Knowledge Proof for L : interactive proof system (P, V) between **prover P** & **verifier V** such that:

- **Completeness & Soundness**
- **Zero-Knowledge:** $\forall$ probabilistic polynomial-time verifier V^* , $\exists$ probabilistic polynomial-time simulator $S \mid \forall x \in L, S(x)$ produces a transcript that is indistinguishable from the real interaction between P and V^* . Symbolically, $\text{View}_{V^*}^P(x) \approx S(x)$
 $\simeq$ The verifier learns nothing beyond the truth of the statement

Source:



Mathematical Definition of Zero-Knowledge Proofs: Concepts and Examples
(by Daisuke Ishii ; 2025)



‘Proof’ in ZKP means :

- (**Prover**, **Verifier**) Interactive (or non-interactive) protocol.

→ **Prover** “demonstrates” knowledge without revealing it.

Probabilistic proof, often repeated multiple times to reduce error probability.

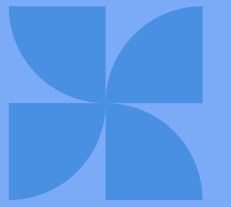
Goal : ensures the 3 foundational properties :

- Completeness, Soundness, Zero-knowledge

Source:



Mathematical Definition of Zero-Knowledge Proofs: Concepts and Examples
(by Daisuke Ishii ; 2025)



EXAMPLES OF ZERO-KNOWLEDGE PROTOCOLS

Hamiltonian Cycle Protocol

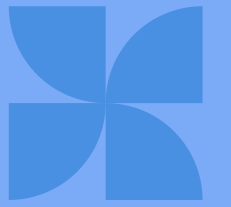
Consider the language:
 $HC = \{ G \mid G \text{ has a Hamiltonian cycle} \}.$

Prover knows a Hamiltonian cycle C in graph G and wants to prove $G \in HC$ without revealing C .

Source:



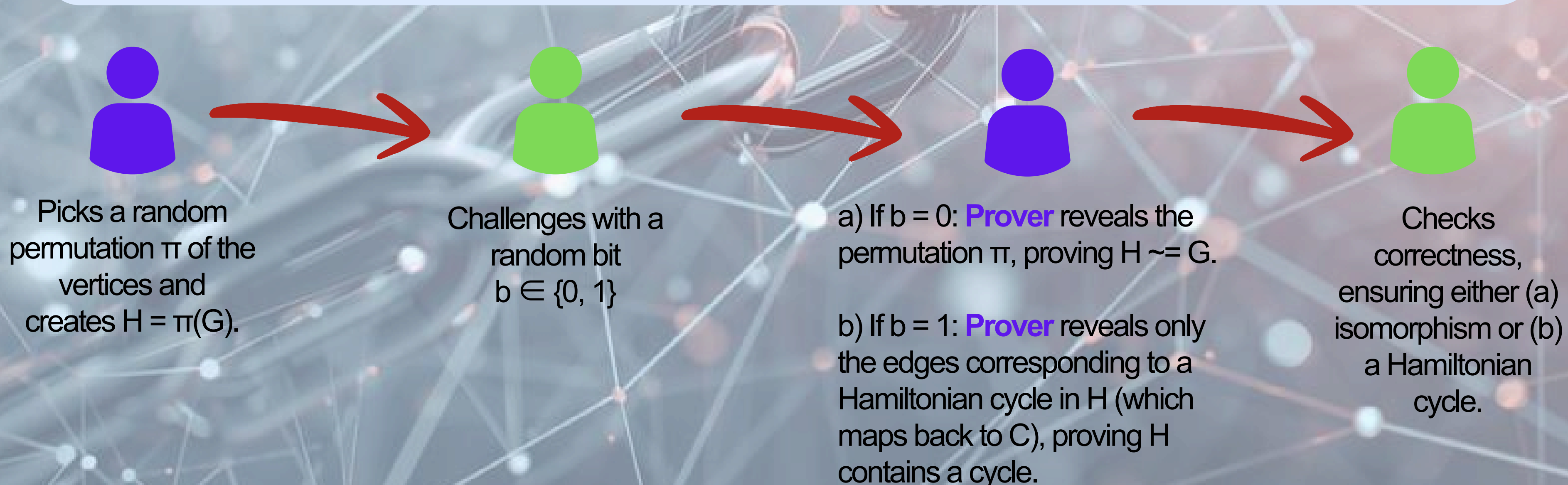
Mathematical Definition of Zero-Knowledge Proofs: Concepts and Examples
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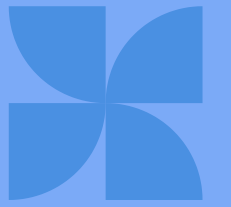


Hamiltonian Cycle Protocol

Consider the language: $HC = \{ G \mid G \text{ has a Hamiltonian cycle} \}$.

Prover knows a Hamiltonian cycle C in graph G . **Prover** wants to prove $G \in HC$ without revealing C .





AN [IMPORTANT] THEOREM

“Every language in NP has a zero-knowledge proof system” under certain conditions*

→ Hamiltonian Cycle is an example of this property.

*if certain cryptographic primitives exist

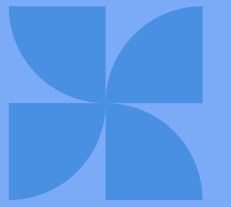
Source:



Proofs that yield nothing but their validity
or all languages in NP have zero-
knowledge proof systems
(by ODED GOLDREICH, SILVIO
MICALI, AVI WIGDERSON ; 1991)

Core idea & model

ZKP programming languages



==>  circum

- Low-level
- Circuit definition language

NOIR

- High-level
- Inspired by RUST

Prover knows a secret → wants to prove $\text{hash}(\text{secret}) = \text{public_hash}$

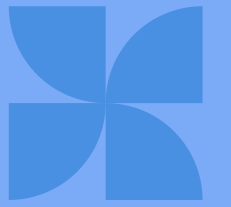
template MysteryFunction() {
 signal input secret;
 signal input publicHash;
 signal output valid;

 valid **<==** (secret * secret) **==** publicHash;
}

template = function
signal = variable
input = given by user
<== = constraint

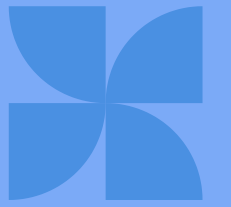
fn main(secret: **Field**, public_hash: **pub Field**) {
 assert(secret * secret == public_hash);
}

fn = function
(pub) Field = (public) variable
assert = constraint



1. **Commitment:** Prover chooses random r and sends $u=g^r$.
2. **Challenge:** Verifier sends a random challenge c .
3. **Response:** Prover sends $z = r + cx$.
4. **Check:** Verifier accepts if $g^z = u * h^c$.

Goal : Prove knowledge of a secret x such that $h=g^x$, without revealing x



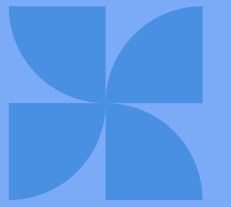
Ali Baba Cave

An interactive demonstration of the zero-knowledge intuition, where João tries to prove he knows the secret word without revealing it.

What is it?

The Ali Baba Cave is the classic example of a zero-knowledge proof. João (Prover) knows the secret word that opens the door inside the cave. He wants to prove this to Thomas (Verifier) without revealing the word.





Schnorr Protocol

A step-by-step demonstration of the Schnorr identification protocol: setup, commitment, challenge, response, and verification.

Schnorr Protocol (simplified)

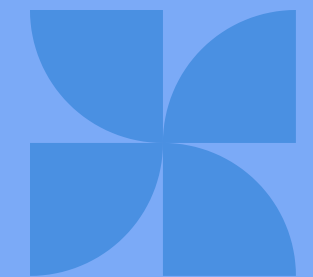
João wants to prove that he knows the secret x such that $g^x \bmod p = y$, without revealing x . This is the core idea behind many modern digital signature systems.

🔒 João: knows x (the secret) | 👤 Thomas: only knows g , p , and $y = g^x \bmod p$



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Thank you for listening



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